

CHOROS

Complete Year Map · Cohort AI Teacher

PHASE 1



Pre-School Setup

July – August

PHASE 2



Crystallization

August

PHASE 3



In-Year Execution

August – May

PHASE 4



Mid-Year Exam

December – January

PHASE 5



End of Year

May – June

SETUP

- Upload calendar teacher

MAP THE YEAR

- Frontier model processes docs frontier

TEACH & LEARN

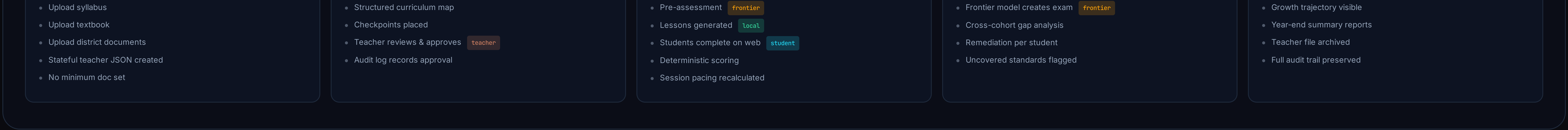
- Roster pulled in

STRONGEST SIGNAL

- Use existing MOY or generate teacher

FULL PICTURE

- Culmination exam



- Documentation as Single Source

Model never goes outside uploaded docs. **Time drifts — content does not.** Full transparency: the system shows exactly what information it has and what is missing.
- Model Tiers

- **Tier 1 (8–14B):** Scoring + gap reports
 - **Tier 2 (32–35B):** Lessons, quizzes, pacing
 - **Tier 3 (70B+):** Crystallization, MOY
 - Build for Tier 2 first.
- Student vs. Non-Student Split

- **Frontier model:** Crystallization, MOY, pre-assessments — batch jobs
 - **Local model:** Lessons, quizzes, misconception tagging — student-facing
 - **Deterministic:** Mastery scoring — never AI
- Audit Log — Non-Negotiable

- Every teacher action recorded
 - Crystallization approved? Logged.
 - Pacing adjusted? Logged with reason.
 - Protects teacher and system.

PHASE 1Pre-School Setup			July – August
INPUTS	WHAT HAPPENS	DECISIONS & PRINCIPLES	
School Calendar Holidays, PD days, early release, assembly dates. ICS file, PDF, or manual entry. Determines available sessions. Syllabus / Scope & Sequence What must be taught and in what order. May include pacing guides, standards references.	Document Ingestion All documents uploaded by teacher. No minimum document set — we work with what's given. If all we get is "Algebra," we produce output grounded in "Algebra" and nothing more. Stateful Teacher JSON Created	► DECIDED No minimum document set. Work with what's given. Document gaps. This is a feature: the system never invents to fill holes.	

Textbook & Supplementary Materials The content the model draws from. PDF, markdown, plain text. Everything the model can reference. District Requirements Testing dates, pacing mandates, district-wide assessments. These become hard checkpoints. State Standards TEKS or equivalent. Deferred beyond POC but the architecture supports standard tagging.	Empty audit log initialized. Teacher identity, course metadata, document manifest recorded. This file grows across the entire year. Calendar Parsing Extract available sessions. Class type documented (45 min daily, 90 min block, A/B schedule, 4×4 block). Total sessions per checkpoint calculated. Visibility Report System shows exactly what information it has AND what it doesn't have. Full transparency on input quality. When someone questions output later, the answer is traceable.	► DECIDED Calendar format is flexible — ICS, PDF, or manual. Extraction approach adapts to input type. ► OPEN QUESTION ICS vs. PDF vs. manual — what's the most common format? Where do we invest parsing effort? ► REALITY CHECK Teachers struggle to document even legally required items. The system cannot solve this — it documents what it receives and makes gaps visible. ► CORE PRINCIPLE Documentation as single source of truth. Time will drift. Content will not. If content is wrong, that's a district-level issue. V1 assumes people do their jobs.
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PHASE 2 Crystallization August		
INPUTS	WHAT HAPPENS	DECISIONS & PRINCIPLES
All Phase 1 Documents Calendar, syllabus, textbook, district requirements, standards. Everything uploaded during setup. Class Type Configuration	Curriculum Map Generation Frontier model processes all documents → structured map. Each checkpoint = a standard or concept cluster with a target date. Map is grounded in source documents only.	► DECIDED Checkpoint dates ideally come from district YAG/WAG/QAG. If not provided, system generates one. If not good enough, fallback to documented adjustment.

45-min daily, 90-min block, A/B schedule, 4×4 block. Determines session counts and lesson pacing.

Frontier Model

Non-student-facing. Batch job. 70B+ recommended for full crystallization; 32B may need chunking or overnight passes.

Checkpoint Placement

Granularity depends on provider quality. Year at a Glance, Quarter at a Glance, or Week at a Glance. District dates become hard checkpoints. If provider gives nothing, system generates a Year at a Glance because checkpoint pacing is basic pedagogy.

Teacher Review & Approval

Teacher reviews the map, adjusts checkpoint dates, approves. Every decision recorded in audit log. System recalculates downstream if checkpoints are moved.

Scope Lock

Once approved, scope is locked. No creeping. No scope expansion without re-crystallization and re-approval.

▸ DECIDED

Class type documented at crystallization — no confusion later about lesson structure.

▸ OPEN QUESTION

Who owns checkpoint placement? Model proposes, teacher approves? What happens to lessons already generated for a moved checkpoint?

▸ OPEN QUESTION

Mid-year transfer student — re-crystallize or just add to existing map? Probably an assessment to place them.

▸ CORE PRINCIPLE

Checkpoints are the drift prevention mechanism. Milestones must be reached at specific times. The model generates lessons that stay inside checkpoint boundaries.

PHASE 3

In-Year Execution

August – May

INPUTS

Crystallized Curriculum Map

Locked scope with checkpoints. The map everything references.

WHAT HAPPENS

Pre-Assessment

Establishes baseline. Can be teacher-provided or frontier-model-generated. Measures growth from this baseline across the year. One per class or per-student diagnostics.

DECISIONS & PRINCIPLES

▸ DECIDED

Student work: webpage path (deep feedback) is preferred. Excel/CSV fallback (shallow — scores only, no misconception signal).

Student Roster Per-classroom. Method TBD for POC. Student identity: number or email. Cross-class identity noted but deferred. Local Model (Lenovo) Student-facing. 32–35B for lesson generation, quiz scoring, misconception tagging. Real-time or near-real-time. Frontier Model Pre-assessment generation only. Batch job at start of year.	Lesson Generation Local model generates lessons bounded by checkpoints. Model must understand class length (45 vs. 90 min) and generate accordingly. Content grounded in uploaded documents only. Student Delivery Individual web pages per student. Quiz with customization: question count, topic focus, question type. Teacher clicks "Quiz," specifies parameters, system produces. Excel/CSV fallback for schools without web access. Deterministic Scoring Right/wrong is math. Never AI. Transparent breakdown shown to learner. Misconception Tagging Local model tags wrong answers with misconception labels. Powers gap reports and remediation. Session Pacing Sessions between checkpoints recalculated dynamically. Holidays, snow days, assemblies absorbed or flagged. Pacing adjustments logged to audit trail.	► DECIDED Teacher interactions are buttons only. No chat. No prompt engineering. The model works behind the scenes. ► DEFERRED Pacing adjustments (lagging / ahead). Document that it can happen, address mechanism later. ► DEFERRED Noise threshold: what triggers teacher notification vs. silent absorption. Too many alerts = ignored. Not enough = not trusted. Discover through testing. ► OPEN QUESTION Frontier vs. local for lesson generation. If frontier, batch job. If local, real-time. Which side? ► OPEN QUESTION What happens when teacher skips a checkpoint entirely? Flag that standards weren't covered, or just log and move on? ► CORE PRINCIPLE Support without judgment. Good teachers get sharper insight. Struggling teachers get stronger scaffolding. No labels, no tiers, no "remedial" flags.
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INPUTS

Crystallized Curriculum Map

What was supposed to be covered by MOY. The validation anchor.

Actual Progress Data

What was actually covered. Checkpoint performance. Pacing adjustments. The gap between plan and reality.

Existing MOY (optional)

District-provided exam. If provided, the system validates it against the curriculum map. Flags items testing uncovered standards.

Frontier Model

Non-student-facing. For MOY generation and validation. Batch job.

WHAT HAPPENS

Path A: Use Existing MOY

District provides exam. Teacher enters results in machine-readable format. System validates exam against curriculum map. Flags items testing uncovered standards. Generates gap analysis.

Path B: Generate MOY

Frontier model creates mid-year exam from what was taught. Item-level validation against original curriculum documents. Teacher reviews every generated item. Audit trail traces every item to a standard.

Cross-Cohort Gap Analysis

Which concepts are breaking down across the class? Not just scores — specific misconception patterns. Which wrong answer → which misunderstanding.

Remediation

Auto-generated remediation bundles per student or per standard. Teacher reviews: Include in Next Lesson or Skip.

DECISIONS & PRINCIPLES

► DECIDED

The MOY is the strongest feedback signal in the system. Pre → checkpoints → MOY → EOY = growth trajectory.

► DECIDED

Uncovered standards must not fail silently. Flagged clearly for teacher awareness.

► OPEN QUESTION

Who sees MOY results? Just teacher, or department head? Privacy boundary.

► OPEN QUESTION

What action does teacher take based on MOY feedback? System suggests pacing changes, or just presents data?

► VALIDATION CONCERN

If the model writes the test AND teaches the lessons, that's a closed loop with no outside check. Teacher review of every generated item is the safeguard. Every item must trace back to a curriculum standard.

► STRATEGIC

MOY generation is where Tier 3 hardware (70B+) becomes transformative. On Tier 2 (32B), teacher-provided MOY with system validation is the realistic path.

PHASE 5

End of Year

May – June

INPUTS

Full Year of Data

Pre-assessment → checkpoint performance → MOY → second-half checkpoints. Every data point since August.

Stateful Teacher File

Grown across the year. Audit logs, pacing decisions, misconception patterns. Potentially large JSON — context window concern.

Student Growth Data

Per-student trajectory. Baseline to final. All misconception history.

WHAT HAPPENS

Culmination Exam (EOY)

Same patterns as MOY. Use existing or generate. Validated against curriculum. Covers full year scope.

Growth Trajectory Report

Pre → Checkpoints → MOY → EOY. Full longitudinal view. Per student and cohort aggregate.

Year-End Summary

What was covered, what was skipped, what standards were mastered, what needs reteaching next year. All grounded in audit trail.

Teacher File Archived

Complete audit trail preserved for professional record. Optional: carry forward to next academic year so the system learns from pacing data.

DECISIONS & PRINCIPLES

► DECIDED

Full audit trail preserved. Professional record. Protects teacher and system.

► OPEN QUESTION

Teacher file carried forward to next year? If teacher teaches same course, system learns from previous pacing data.

► OPEN QUESTION

Teacher file size by June — can the model still load it into context? Chunking strategy needed?

► CORE PRINCIPLE

The purpose of the year-end is not to judge — it's to document. What happened, what was learned, what carries forward. The audit trail is the artifact.

● Teacher

Uploads docs, reviews & approves, adjusts pacing. Buttons only. Never chats with the model. Every decision recorded in audit log.

● Frontier Model

Non-student-facing. Crystallization, MOY generation, pre-assessments. Batch jobs — never real-time. 70B+ ideal; 32B with chunking.

● Local Model (Lenovo)

Student-facing. Lesson generation, quiz scoring, misconception tagging. Real-time or near-real-time. 32–35B on local hardware.

● Student

Completes lessons on individual web pages. Deterministic scoring + LLM misconception feedback. Excel/CSV fallback for schools without web access.